

Application Architecture Diagram

Overdrive

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Summary

OverDrive – Application Architecture Diagram

1-Data sources (Providers)

2- Data Ingestion & Normalization

3- Backend OverDrive

4- Databases and storage

5- Client applications

1- VR Application (Meta Quest – Unity)

2- Mobile application (partner)

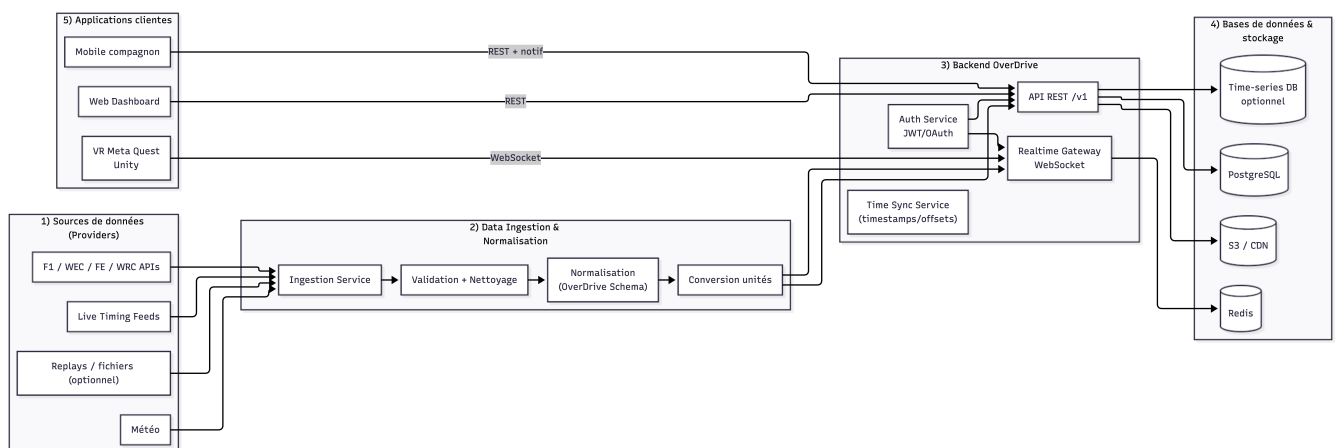
3- Web Dashboard (management + analysis)

6- Conclusion

Overview of the architecture

The overall functioning of the architecture can be summarized as follows :

- Data from various championships is collected from multiple sources.
- It is then standardized in a common schema, ensuring consistent processing.
- A central backend handles authentication, profile management, and data distribution via API and WebSocket.
- Databases are specialized according to their use cases (persistence, real-time, storage).
- VR, mobile, and web clients consume this data according to their functional role.



1- Data sources (Providers)

The first layer of the architecture corresponds to the different data sources related to motorsport championships.

This data may come from :

- Public or partner APIs (F1, WEC, FE, WRC, etc.),
- Live timing feeds,
- Weather data,
- Replay files (optional, for offline analysis).

This approach makes it possible not to depend on a single supplier and facilitates the future integration of new championships or data sources.

2- Data Ingestion & Normalization

The raw data from the different providers are heterogeneous by nature. They therefore pass through a layer dedicated to ingestion and normalization.

This layer ensures :

- Secure collection of data streams,
- Validation of received data,
- Cleaning of any inconsistencies,
- Conversion to a unified internal schema: the OverDrive Schema,
- Standardization of units (km/h, °C, seconds, etc.).

This standardization ensures that all client applications receive structured, consistent, and actionable data, regardless of the original championship.

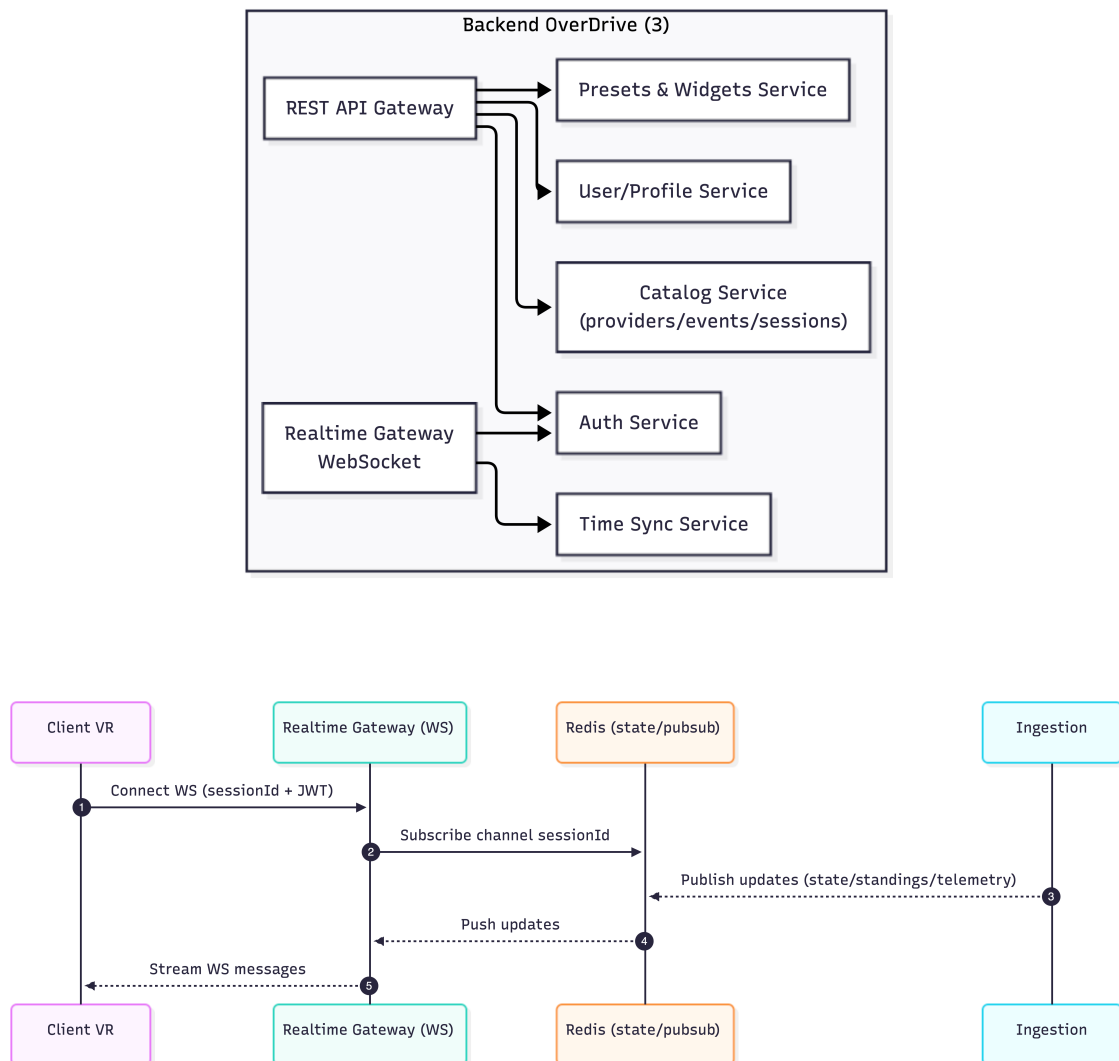


3- Backend OverDrive

The backend is the functional and technical core of the system. He performs several critical roles :

- User authentication and management (account creation, secure login, OAuth).
- Management of user profiles, presets, and custom layouts.
- Exposure of a REST API for mobile and web applications.
- Real-time data distribution via a WebSocket-based Realtime Gateway.
- Time synchronization service, ensuring data consistency through timestamp and offset management.

This architecture allows fluid, secure and efficient communication between the backend and the different clients.

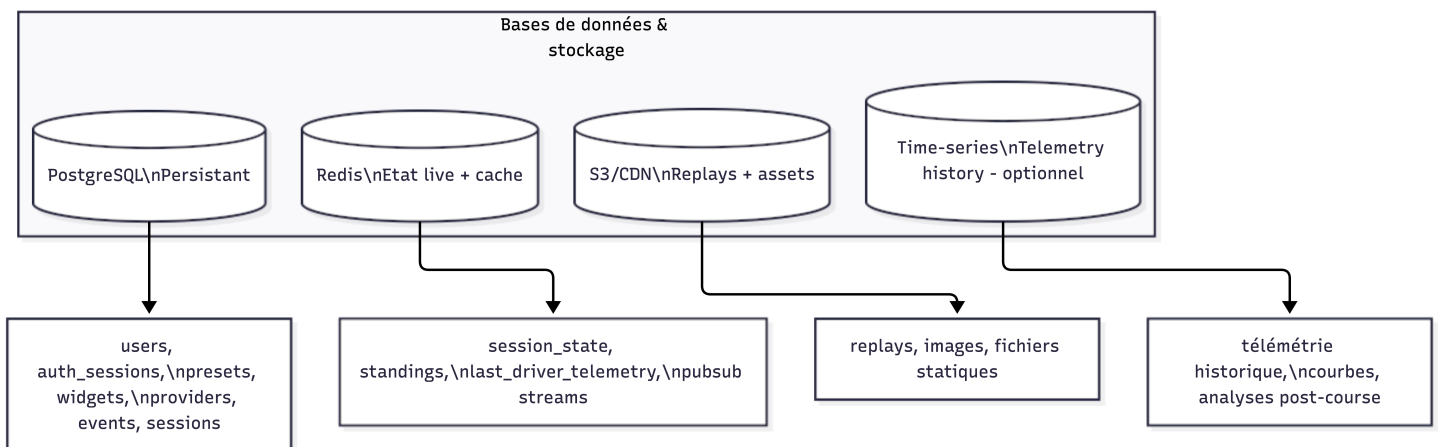


4- Databases and storage

OverDrive relies on several storage solutions, each adapted to a specific data type :

- **PostgreSQL**
 - Used for persistent data: user accounts, profiles, presets, configurations.
- **Redis**
 - Used as a cache and for real-time state management, to optimize latency and performance.
- **Stockage object (S3 / CDN)**
 - Intended for replays, assets and static files.
- **Time-series database (optional)**
 - Used for telemetry history and advanced performance analysis.

This separation improves both the **scalability** and the **resilience** of the system.



5- Client applications

The data processed and distributed by the backend is consumed by three types of complementary clients.

VR application (Meta Quest – Unity)

The VR app is the immersive heart of the OverDrive experience. It allows:

- Displaying data as spatial widgets in a virtual environment,
- Direct user interaction with their personalized interface,
- Receiving real-time data via WebSocket.

It offers a premium, immersive and differentiating experience.

Mobile application

The mobile application acts as a companion application.

It allows:

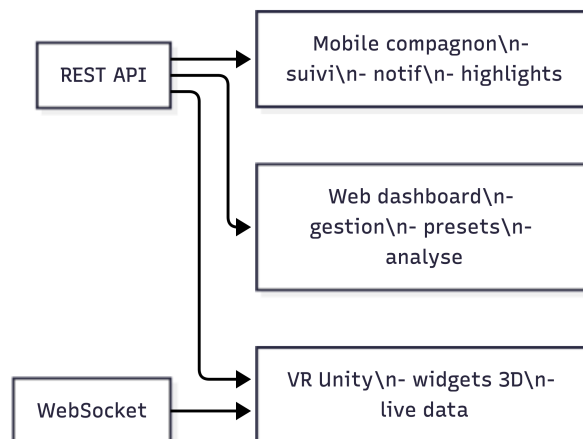
- the complementary follow-up of a race,
- the receipt of notifications,
- consultation of highlights and key data,
- rapid access to mobile information.

Web Dashboard

The Web Dashboard plays a more administrative and analytical role.

It allows :

- management of user accounts,
- the setting of layouts and presets,
- access to replays and historical data.



Conclusion

The architecture of OverDrive is based on a modular, robust and scalable approach.

It allows :

- The future addition of new championships and data sources,
- A clear separation of technical responsibilities,
- High-performance real-time data distribution,
- Native compatibility with VR, mobile, and web clients.

This architecture constitutes a solid technical basis, capable of supporting the Premium ambitions of OverDrive and its long-term evolution.